

Project 1

Cahier des charges / mission statement

- Compare the drag and the downforce of 3 configurations of a car
 1. base: without rearwing or diffuser
 2. with rearwing
 3. with diffuser
- Measure each configuration at 3 different speeds
 1. 80 km/h
 2. 120 km/h
 3. 160 km/h
- The body of the car is free to choose
- no measurement or studies of the thermal parts, of the engine or electric parts of the car
- at the end of the project, im going to deliver
 1. A pdf report of all the decisions, calculations and other steps to make this project work.
 2. All the codes used to be able to do the simulations.
 3. The meshes used to do the approximations
- I will consider this project a success if
 1. The results are reliable: difference $< 5\%$ between 2 meshes
 2. Realists: the results i obtained are in the norm of what's happening right now
 3. significative difference: the difference between 2 configurations is bigger than the uncertainty.

Planning:

Today 30/09: finish mission statement and planning

Week 1: 01/10 → 07/10: learning about Github, how to document a project, README, a good report

- Detailing the middle steps of the project
 - What software im going to use
 - What are the first steps
 - What body / base configuration am i going to use?
 - do i need / want to design the rearwing, diffuser or whole body by myself?

Week 2 - 5: 08/10 → 05/11: project body of work

→ Details TBD during week 1.

Week 6: creation of the report (goal between 15-25 pages pdf).

- 05/11
 - Introduction
- 12/11
 - Methodology and theory used
 - first step of the project
 - second step
 - results
 - Conclusion

Week 7: publication 12/11 → 19/11.

- linkedin
- portfolio creation
- youtube video?

How much time will i allocate to this project?

Considering my studies, sports activities and personal obligations, i think that i can realistically work for 4 hours / week on this project.

Those 4 hours are going to be distributed on two blocks of 1h30 during the week and one block of one hour to prepare, organize and document during the weekend.